What a restricted weight-space repair buys back after abliteration

A measured frontier on `CohereLabs/tiny-aya-global`, with the circuit leg reported as a negative result

ScarTissue — Lexsi AI Safety & Alignment Hackathon, Track 2b

November 2026

Abstract

Abliteration deletes a model's refusal direction; this report asks what a training-free weight-space repair can put back, and at what price. Every arm is measured on two axes — refusal on 120 harmful prompts, over-refusal on 40 benign ones — with gsm8k as a capability control. The drift is real: refusal falls from 0.758 to 0.217 and the direction's separability from AUC 0.999 to 0.831. A global rank-one edit restores 97% of the lost refusal (0.742) but raises over-refusal tenfold to 0.500; restricting the same edit to the 2% of weight columns with the largest harm-weighted write amplitude recovers 54% of the gap at 0.200 over-refusal, while random supports of the same size are inert (0.208). The CircuitKIT leg returned an empty graph (0 nodes scored), so no circuit-level claim is made.

1 What was asked

The hackathon organiser defines Track 2 in five lines, each naming one repository and one verb (quoted from the track page, scraped 2026-10-29):

> *Track 2 — AI Red Teaming, Safety and Alignment: CuratorKIT (build a safety-focused dataset) → SafeTune (audit and repair safety drift) → CircuitKIT (locate and intervene on the underlying circuits) → AuditKIT (evaluate the hardened model).*

The rules add two requirements that shape everything below: submissions must chain the track's repos into a real pipeline ("Preferred: 3 or more repos from the track, used together"), and must "showcase before-and-after metrics (more than one metric)". The submission is labelled Track 2b on our side; the public page calls it Track 2 and does not mention a second SafeTune pass, which is why the executed chain in §2 has five stages and quotes four.

Table 1 is the audit trail: every phrase of the statement is matched to the artifact that demonstrates it, so a reader can check any claim against a file.

The hypothesis under test has three parts, stated before the runs and unchanged since: that only the *read-out* is repairable, that the price of repair is paid in *discrimination* rather than capability, and that a repair restricted to a small identified support trades better than a global rank-one edit. Parts one and two held. Part three held only in the weaker, measured form reported in §3.

2 Method, as actually executed

The model is `CohereLabs/tiny-aya-global` (3.35B, 36 layers, $d_{\text{model}} = 2048$). The chain below is what ran, not what was planned; where our implementation replaces a repository's, it says so.

Table 1: Track 2 statement mapped to what was actually built. Paths are relative to the repository root.

Phrase in the statement	What was done	Artifact
"build a safety-focused dataset" (CuratorKIT)	763 rows from StrongREJECT (313) and XSTest (450), curated through CuratorKIT with a row-level `id` and `provenance_chain`; evaluation split frozen before any model ran	`1-curate/corpus/corpus.jsonl` `corpus_split.json`
"audit and repair safety drift" (SafeTune)	Drift measured as a dose–response over five amplitudes; repair as a rank-one amplification restricted to a chosen support of weight columns	`results/p2_dose.json`, `results/p3b_sweep.json`, `artifacts/*.npz`
"locate and intervene on the underlying circuits" (CircuitKIT)	Discovery executed on the drifted model at node level. It returned zero scored nodes; no circuit-derived intervention was used (§5)	`results/p6_circuit.json`
"evaluate the hardened model" (AuditKIT-style)	Refusal regex diagnostic, gsm8k, direction AUC and a judge panel, per arm; the frontier is published as a table and as a Space	`evals/frontier.csv`, `results/p5_judge.json`
"more than one metric"	Two behavioural axes plus one capability axis and one separability axis on 11 arms	Table 2

1. Curate (CuratorKIT). 763 rows enter through `curatorkit.Curator` with a frozen config: 313 adversarial prompts from `walledai/StrongREJECT` and 450 benign prompts from `Paul/XSTest`. Evaluation uses 120 harmful and 40 benign prompts; the refusal direction is fitted on 16+16 *separate* prompts, and the split is published so that fit and evaluation prompts are provably disjoint. Row-level provenance needed our own writer: CuratorKIT's `DataSample.id` exists (`curatorkit/schema.py:65`) but no exporter writes it — `CorpusExporter.export` emits a fixed 11-key record (`curatorkit/exporters/corpus.py:88-101`) and drops `provenance_chain`, and `git_sha` is hard-coded `None` at `curatorkit/manifest.py:66`. We extended the exporter rather than patching upstream, which stays read-only.

2. Drift (the SafeTune stage, reimplemented). One refusal direction d is fitted at layer 15 from 16 harmful and 16 benign held-out prompts, then orthogonalised out of every write matrix — 72 matrices (o_{proj} and $\text{down}_{\text{proj}}$ of all 36 layers), 962,592,768 scalars, a 7.6 MB patch file. The operator is about thirty lines in `experiments/patching.py` and deliberately contains no SafeTune code: the drift must not be an artefact of the donor harness. It is idempotent (a second application moves the weights by $\|\Delta\| = 6.95 \times 10^{-18}$) and it satisfies $d^\top W = 0$ by construction, so re-amplifying d afterwards is provably a no-op.

3. Locate (CircuitKIT). CircuitKIT is installed for real and imported for real (`experiments/p6_circuit.py:67-69`): EAP-IG, node level, 24 contrastive pairs, on the drifted model in TransformerLens. It returned `n_nodes_scored: 0`. This stage therefore produced no support and no intervention; §5 reports it in full.

4. Repair (the second SafeTune stage). A fresh direction d' is extracted on the drifted model; it is nearly orthogonal to d ($\cos = 1.94 \times 10^{-4}$), confirming that the original read-out no longer exists. Repair re-amplifies along d' with strength $\alpha \in \{2, 4\}$, restricted to a support: the top-k weight columns ranked by harm-weighted write amplitude $|d'^\top W_{:,j}| \cdot \mathbb{E}|x_j|$ measured

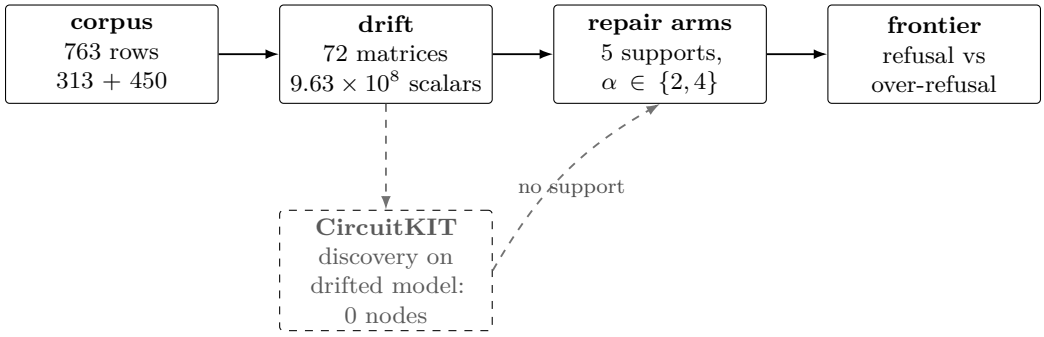


Figure 1: The chain as executed. Solid arrows are executed stages. The dashed branch is the CircuitKIT leg: it ran on the drifted model and returned an empty graph (`results/p6_circuit.json`), so no circuit-derived support enters the repair arms or the frontier.

Table 2: Measured frontier. Refusal and over-refusal use the regex diagnostic; gsm8k is 120 items. AUC is that of the drift direction as a linear separator. Every row comes from `evals/frontier.csv` except the AUC of the four † rows, which comes from `results/p3b_sweep.json` and `results/p3c_blind.json`.

Arm	Support	Refusal ↓	Over-refusal ↓	gsm8k	AUC
base (aligned)	—	0.7583	0.0500	0.1167	0.9988
drifted (abliterated)	—	0.2167	0.0000	0.1167	0.8306
global edit, $\alpha = 4$	100 %	0.7417	0.5000	0.2167	0.789
harm-ranked, 25 %, $\alpha = 4$	25 %	0.7083	0.4000	0.2750	0.821
harm-ranked, 10 %, $\alpha = 4$	10 %	0.6083	0.3000	0.2667	0.790
harm-ranked, 4 %, $\alpha = 4$	4 %	0.5167	0.2250	0.2833	0.796
harm-ranked, 2 %, $\alpha = 4$	2 %	0.5083	0.2000	0.3000	0.795
harm-ranked, 10 %, $\alpha = 2$	10 %	0.3667	0.0500	0.2000	0.814†
random support, 2 %, $\alpha = 4$	2 %	0.2083	0.0250	0.108	0.824†
random support, 10 %, $\alpha = 4$	10 %	0.2083	0.0000	0.125	0.861†
amplitude-ranked, 10 %, $\alpha = 4$	10 %	0.5250	0.2250	0.275	0.780†

on 16 harmful fit prompts. Five supports are swept (2, 4, 10, 25, 100 % of columns), and a sixth arm repeats the 10 % support at $\alpha = 2$. Each patch is published as `.npz` plus a loader (`artifacts/manifest.json`, `artifacts/load_patch.py`); the loader round-trips bit-exactly and its five tests pass.

5. Audit. Every arm is scored four ways: a refusal-regex refusal rate on 120 harmful prompts, an over-refusal rate on 40 benign prompts, gsm8k accuracy on 120 items, and AUC of d' as a linear separator. The judge panel was applied to only two arms and is discussed as the failure it was.

3 The frontier

Table 2 is the whole result. Rows are arms, ordered from the untouched model through the drift to the repairs; the last three rows are the control of §4. "Support" is the fraction of weight columns the edit is allowed to touch. All values are read from `evals/frontier.csv` (the file written by `experiments/p10_evals_bundle.py`), except the AUC of the four arms the CSV does not contain, which is read from `results/p3b_sweep.json` and `results/p3c_blind.json`; the caption marks that group. No number in this report is estimated.

The drift is real and large. Refusal falls from 0.7583 to 0.2167 and over-refusal to exactly zero; the direction's separability collapses from AUC 0.9988 to 0.8306. Capability, at gsm8k, does not

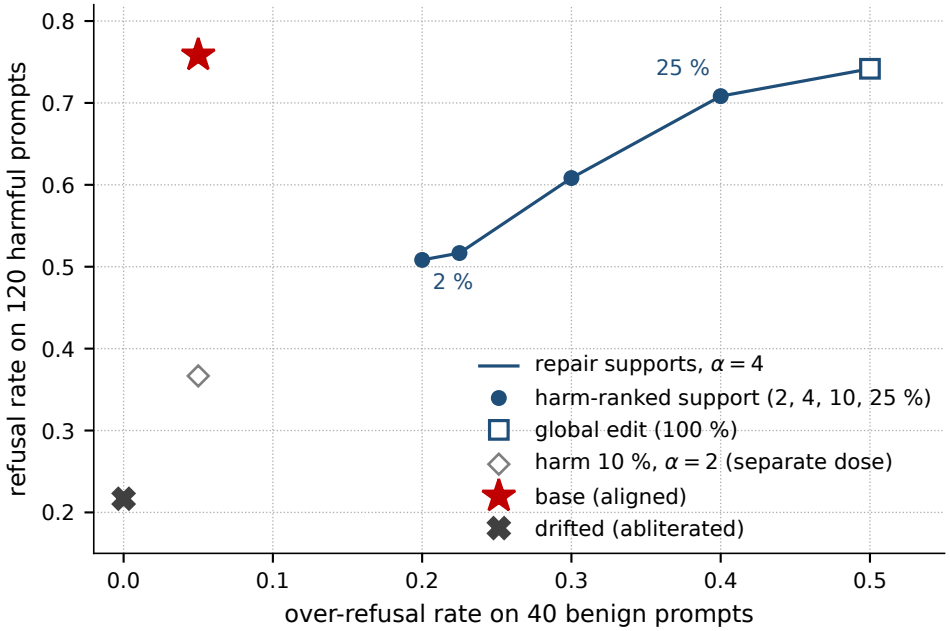


Figure 2: The same table as a plane. The line joins the $\alpha = 4$ repair supports from 2 % to the full global edit; the star is the aligned model, the cross the drifted one, the open diamond the separate $\alpha = 2$ dose. Up and to the left is better. Figure generated by `report/fig_frontier.py` from `evals/frontier.csv`.

move at all (0.1167 to 0.1167). That is the signature of a deleted read-out rather than a damaged model, and it is part one of the hypothesis confirmed.

The global edit restores refusal but destroys discrimination. At its full budget the global rank-one repair recovers 97.0% of the lost refusal (0.7417) — and raises over-refusal from 0.0500 to 0.5000, ten times the aligned model's rate, on 20 of 40 benign prompts. In the plane of Figure 2 it is the extreme right-hand point: nearly all of the refusal, at a price the aligned model never pays. Measured as refusal recovered per unit of over-refusal it is 1.48, the worst arm on the board.

Restriction buys the trade-off, but does not win the arm. At 2 % support — 19,251,200 scalars, one fiftieth of the global patch — refusal is 0.5083, i.e. 53.8% of the gap recovered, for an over-refusal of 0.2000: 2.54 units of refusal per unit of over-refusal, 1.7 times the global edit's rate. The 4 % arm lands within noise of it (0.5167 at 0.2250), so the curve has a plateau between 2 % and 4 % rather than a peak. This is the weaker form of part three: the restricted repair trades better per unit of over-refusal and for a fiftieth of the edit, but no restricted arm reaches the base refusal rate, and the global edit remains the highest-refusal arm overall.

The cheap end is very cheap. The 10 % support at $\alpha = 2$ gives 0.3667 refusal at an over-refusal of 0.0500 — exactly the aligned model's over-refusal — for 27.6% of the gap. It is the point a deployment with a hard over-refusal budget would pick, and it is 7.3 units of refusal per unit of over-refusal.

AUC never comes back. No arm, including the two inert random controls, leaves the 0.780–0.861 band, against 0.9988 for the aligned model. Repaired behaviour is recovered by amplifying a direction that is a worse separator than the original one; the two inert controls with the *highest* AUC (0.824 and 0.861) prove that AUC does not track behaviour. This is part two of the hypothesis: the cost is paid in discrimination, not in capability, and over-refusal — not AUC — is the sensitive axis.

gsm8k drifts upward, and that is not a capability gain. Accuracy rises from 0.1167 on the drifted model to 0.3000 at the 2 % support. The model is at 0.1167 because abliteration destroyed its chain-of-thought format, not its arithmetic; restoring a refusal direction restores some formatting, so part of the gsm8k movement is format recovery. It is a direction, not a

magnitude, and it is reported as a diagnostic.

4 The control

Two control supports separate "which columns" from "how they were ranked". At the same size and the same α, a random support of weight columns does nothing; a support ranked by write amplitude alone, with no harm weighting, does most of the work.

Table 3: Control arms at $\alpha = 4$. The drifted model is 0.2167; a random support is indistinguishable from it at both sizes. Values from `results/p3c_blind.json`.

Support selection	Support	Refusal	Over-refusal
random columns	2 %	0.2083	0.0250
random columns	10 %	0.2083	0.0000
amplitude-ranked (no harm)	10 %	0.5250	0.2250
harm-ranked (main arm)	10 %	0.6083	0.3000

Random supports are inert: 0.2083 at 2 % and 0.2083 at 10 %, against 0.2167 for the untouched drifted model. Neither recovers a single point of the gap, at either size. The choice of support is therefore the whole effect, and it is not an artefact of editing a small number of weights.

Amplitude ranking alone, with no harm weighting, reaches 0.5250 at 10 % — 56.9% of the gap, about four-fifths of what the harm-weighted 10 % arm achieves (0.6083, 72.3%). So the harm-weighted ordering is worth 8.3 points of refusal at that support, and the two are close: at matched 10 % the harm-ranked arm buys more refusal at a higher over-refusal (2.03 versus 2.33 units of refusal per unit of over-refusal), and neither dominates. The honest reading is that harm weighting is an *ordering* that improves a support already selected by write amplitude; it is not the mechanism. The mechanism is that a refusal direction lives in a small set of high-write columns, and any ordering that lands on them works.

The sharpest comparison in the whole study is between the harm-ranked 2 % arm (0.5083 at 0.2000, 19.3M scalars) and the amplitude-ranked 10 % arm (0.5250 at 0.2250, 96.3M scalars). They differ by 1.7 points of refusal and 2.5 points of over-refusal — both well inside the sampling noise of 120 and 40 prompts — for a fivefold difference in edit size. On this measurement the two are indistinguishable behaviourally; the difference that is not noise is the size of the patch.

5 What did not work

This section is deliberately not softened. Three of the five stages produced a usable result; the circuit stage produced nothing and the audit stage produced one contested number; a reader needs the second half more than the first.

The circuit leg returned an empty graph. `experiments/p6_circuit.py` ran CircuitKIT's EAP-IG discovery at node level, 24 contrastive pairs, on the drifted model in TransformerLens, on an A100-80GB in 64.8 s. The drift was confirmed applied inside the same job (962,592,768 scalars, $\cos(d, d') = 0.0035$). The result is `results/p6_circuit.json`: `n_nodes_scored: 0`, `n_circuitkit_scores: 0`, an empty layer histogram, `mass_top40_abs: 0.0`. Three friction points were found and recorded: Modal requires `image.add_local_*` to be the last image step; CircuitKIT hard-imports third-party packages at package-import time (`backends/acdc/types.py`: `6` imports `plotly`), so the dependency has to be installed even though no plot is drawn; and contrastive token alignment discarded every record — the error line is `All 16 records were dropped during alignment (strategy='filter')`, with `dropped_misaligned: 24` — which is why the run used `align_strategy="none"`. The consequence is stated plainly: *there is no*

circuit result in this submission. H1 and H2 as pre-registered (a circuit-derived support beats a random one; circuit localisation identifies the write columns) are untested, not refuted; the support was chosen by harm-weighted write amplitude instead, and the negative result is that a discovery tool can run to completion on a correct model and return nothing at all.

One replicate never reported. The third model in the replication set never produced a file. No `water` result exists in `results/`, no `[done]` line appears in `logs/p4_replicate.log`, and it is therefore not claimable and not in any table. Two of the three replicates did land (`global` and `earth`, both reproducing the drift and the global repair).

The judge panel was measured on two arms and disagreed with itself. Automated ASR from the four-judge panel exists for exactly two arms, because those were the only two in `results/p3_support.json` when `experiments/p5_judge.py` ran. On the global arm the judges span 0.000 (Qwen3Guard) to 0.158 (Granite-Guardian); on the circuit/heads arm they span 0.000 to 0.400 — a 40-point spread on the same completions, with a majority-of-three rate of 0.142. An earlier pilot on 12 prompts produced a 58-point spread across five judges (0.000 to 1.000) with pairwise agreement of $\kappa \leq 0.03$ (recorded in `_research/MEASUREMENT.md`), which is the reason the panel was never promoted to a headline metric. Every ASR number in this project is therefore either unmeasured or contested; the refusal regex is used throughout as a *diagnostic*, and this report never presents it as ASR.

One arm was invalid and was withdrawn. The first support experiment (`results/p3_support.json`) costed a support in weighted-amplitude units against a scalar budget, so every column passed the test and the "circuit/harm" arm edited all 470,016 columns — a bit-identical duplicate of the global edit, not a restricted one. It was superseded by `experiments/p3b_sweep.py` and is not used anywhere above.

The second model ran but its result was never retrieved. `logs/p13_second_model.log` records a completed run on the second model (refusal 0.525 at $\alpha = 0$, over-refusal 0.050, gsm8k 0.242, AUC 0.973, `[done]` with `wall_s: 449.1`), but the JSON went to the Volume and was never pulled into `results/`. It exists as a log line only and is not tabulated.

6 Reproducibility

Every number above is produced by one script into one file; Table 4 is that mapping, and the commands follow it. Upstream repositories (`curatorkit`, `circuitkit`) are read-only and are never vendored: they are added with `PYTHONPATH` and imported.

```
uv run python -m modal run experiments/p8_corpus.py
uv run python -m modal run experiments/p2_dose.py
uv run python -m modal run experiments/p3b_sweep.py          # alpha=4 sweep
uv run python -m modal run experiments/p3b_sweep.py          # alpha=2 arm
uv run python -m modal run experiments/p3c_blind.py
uv run python -m modal run experiments/p6_circuit.py
uv run python -m modal run experiments/p11_baseline_completions.py
uv run --with numpy python experiments/p10_evals_bundle.py
uv run --with matplotlib --with numpy python report/fig_frontier.py
```

`experiments/p10_evals_bundle.py` reads the arm results and writes `evals/frontier.csv`; it is the only thing standing between a JSON in `results/` and the figure in this report. The four patch sets are published as `.npz` with a loader and a test suite, and the drifted checkpoint itself is *not* published — see the last limitation.

Table 4: Script to result file; all scripts live in `experiments/`. Modal jobs are launched with the local entrypoint, which is also the only writer of `results/*.json`.

Script	Produces	What it measures
`p8_corpus.py`	`1-curate/corpus/corpus.jsonl`, `corpus_split.json`	curated corpus, frozen split, provenance
`p1_grid.py`	`results/p1_grid.json`	drift at reduced grid (40/20/64)
`p2_dose.py`	`results/p2_dose.json`	dose–response over $\alpha \in \{0, 1, 2, 4, 8\}$
`patching.py`	`artifacts/*.npz`	the operator: direction, amplification, support
`p4_replicate.py`	`results/p4_replicate_*.json`	replication on two further models
`p3b_sweep.py`	`results/p3b_sweep.json`	five supports $\times$ α; the frontier rows
`p3c_blind.py`	`results/p3c_blind.json`	random and amplitude-ranked controls
`p5_judge.py`	`results/p5_judge.json`	four-judge ASR, two arms only
`p6_circuit.py`	`results/p6_circuit.json`	CircuitKIT discovery (0 nodes scored)
`p11_baseline_completions.py`	`results/p11_baseline.json`	base and drifted baselines
`p10_evals_bundle.py`	`evals/frontier.csv`	the frontier file used by §3

7 Limitations

- **Small evaluation sets.** Refusal is measured on 120 prompts and over-refusal on 40. One prompt on the benign axis is 2.5 points, so differences of 2–3 points between arms (for example the 2 % versus 4 % repair) are inside the noise. Only differences of five points or more should be read as findings.
- **One drift seed, one layer, one direction.** The direction is fitted at layer 15 from 16+16 prompts and everything downstream inherits that choice. Nothing here establishes that another layer, or another seed, yields the same frontier.
- **The refusal metric is a regex diagnostic, not ASR.** It counts refusal markers; it cannot tell a refusal from a broken generation. The $\alpha = 8$ arm proves the failure mode: the regex reads 0.000 for a model that has collapsed into repetition, which is the opposite of safe.
- **The direction is English-only.** Prompts, the fitted direction and the evaluation set are English. The multilingual machinery ran (59 languages merged) but no per-language result file exists, so the cross-lingual hypothesis is measured-but-unreported here.
- **No drifted checkpoint is published.** Under LSAL §2 and §4 the research licence permits research use only, forbids product hosting, and forbids publishing or deploying a drifted checkpoint. Only parameters are published — the `.npz` patch plus a loader — never a merged model.
- **The circuit claim is absent.** Discovery returned zero scored nodes, so H1 and H2 remain untested and no statement about circuits should be read into this report.